

1 Introduction

1.1 Definetly include

- Masting has different hypotheses
- Different hypotheses concentrating at different stages
- Linking traits-hypotheses-masting can help us better understand what has been selected on for masting species
- How different traits are related to masting

1.2 Maybe include

2 Discussion

2.1 Definetly include

- Masting and seed weight (Previous discussion on whether big seeds or small seeds should be link to masting)
- Masting pattern in gymnosperm and angiosperm, gymnosperm are more likely to be masting species
- Better definition of masting
- Masting is a spectrum
- Masting is strongly related to environment as well, so traits alone might not be able to explain
- Flower masting and fruit masting, different hypotheses might act.
- Look for patterns within the clade can be more informative, oak.

2.2 Maybe include

- Masting is a spectrum, difficult to define masting.

3 Results

3.1 Gathered data

- Number of data available for angiosperm and gymnosperm originally from book, with masting data, and the structure of data
- Angiosperm (124, 98, Y/N: (51/47); Gymnosperm (64, 61, Y/N: (57/7))
- Strong masting more common for gymnosperm

- Number of traits we have for different hypotheses for angiosperm and gymnosperm
- For angiosperm, we have data for five traits (seed weight, fruit size, oil content, seed dispersal mode, seed dormancy) related to predation satiation hypothesis; two traits (pollination mode, reproductive type) related to pollination coupling; two traits (drought tolerance, leaf longevity) related to resource matching hypothesis. For gymnosperm, we only have oil content for masting species and all the gymnosperm species are wind pollinated.

3.2 Results for each hypothesis

- Predation satiation: we found only seed weight is related to masting with having bigger seeds will increase the chance of being a strong masting species, however, the log-odds is only 0.2. We also only found this in angiosperm group.
- Pollination satiation: we found wind pollination is related to masting for angiosperm species, the log-odds for wind compare to animal or both are 1.72 and 1.23 respectively. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of masting species.
- Resource matching: We don't find any traits related to resource matching hypothesis have relationship with the masting pattern.

3.3 Multidimension-NMDS

- We don't see masting aligned well along any axis in the NMDS analysis.

3.4 Phylogenetic patterns

- First show the phylogenetic signal for masting. Masting has a strong phylogenetic signal for gymnosperm, and no phylogenetic signal for angiosperm
- Show which traits have the strongest phylogenetic pattern, (it may not be fully comparable though, not only because the way lambda is calculated for continuous traits and categorical traits are different, but also because lambda might depend on the data structure when sampling is uneven, take leaf longevity as an example).
- seed weight and pollination mode both have strong phylogenetic signal, (it is hard to tease out these traits from masting because of the phylogenetic signal, limit the inference)
- Data is strongly nested in phylogeny, and data is sparse. We have to assume the sampling is even.